Lab 2

6. Create a view for student names with their subjects' names which they will study.

```
postgres=# CREATE VIEW stu_sub_view AS
SELECT student.e_name, subject.sub_name
FROM student
JOIN stu_sub ON student.id = stu_sub.stu_id
JOIN subject ON stu_sub.sub_id = subject.id;
CREATE VIEW
postgres=# select * from stu_sub_view;
 e_name  |  sub_name
---------+-------------
 Mohamed | python
 Ahmed   | java script
 Omar    | database
(3 rows)

postgres=#
```

7. Create a view for tracks names and the subjects which belong to it.

```
postgres=# CREATE VIEW track_sub_view AS
SELECT t.track_name, sub.sub_name
FROM track t
join track_sub ts on t.id = ts.track_id
join subject sub on ts.sub_id = sub.id;
CREATE VIEW
postgres=# select * from track_sub_view;
 track_name  |  sub_name
-------------+-------------
 fullstack   | python
 mern        | java script
 open source | database
(3 rows)

postgres=#
```

Lab 3

1- Write a query to find out which subjects are not associated with any track.

```
postgres=# INSERT INTO subject (sub_name,max_score) VALUES ('jQuery','90');
INSERT 0 1
postgres=# SELECT sub_name
postgres-# FROM subject
postgres-# WHERE id NOT IN (
postgres(#     SELECT sub_id FROM track_sub
postgres(# );
 sub_name
----------
 jQuery
(1 row)

postgres=# 
```

==

2- Display name and age of each students.

```
postgres=# SELECT e_name, EXTRACT(YEAR FROM AGE(CURRENT_DATE, birth_date))
FROM student;
 e_name  | date_part
---------+-----------
 Mohamed |        24
 Ahmed   |        27
 Omar    |        17
(3 rows)

postgres=# 
```

3- Display the name of students with their rounded score in each subject

```
postgres=# SELECT s.e_name,ROUND(g.grade),sub.sub_name
FROM student s
join grades g ON s.id = g.stu_id
postgres-# join subject sub ON g.sub_id = sub.id;
 e_name  | round |  sub_name
---------+-------+-------------
 Mohamed |    85 | python
 Ahmed   |    88 | java script
 Omar    |    92 | database
(3 rows)

postgres=#
```

4- Display the name of students with the year of Birthdate.

```
postgres=# SELECT e_name, EXTRACT(YEAR FROM birth_date)
FROM student;
 e_name  | date_part
---------+-----------
 Mohamed |      2001
 Ahmed   |      1998
 Omar    |      2007
(3 rows)

postgres=#
```

===

5- Add new exam result, in date column use NOW() function.

```
postgres=# INSERT INTO exam (date)
VALUES (NOW());
INSERT 0 1
postgres=# select date from exam;
    date
------------
 2025-04-29
 2025-06-15
 2025-04-30
(3 rows)

postgres=#
```

6- Write a query to calculate the average grade obtained by a specific student across all exams.

```
postgres=# insert into grades (stu_id, sub_id, exam_id, grade) VALUES (1,3,2,74);
INSERT 0 1
postgres=# select AVG(grade)
from grades
where stu_id = 1;
         avg
---------------------
 79.5000000000000000
(1 row)
```

7- Write a query to replace all occurrences of 'gmail.com' in email addresses with 'iti.com'.

```
postgres=# UPDATE student
postgres-# SET email = LEFT(email, LENGTH(email) - 9) || 'iti.com'
postgres-# WHERE RIGHT(email, 9) = 'gmail.com';
UPDATE 3
postgres=# select email from student;
      email
-----------------
 mohamed@iti.com
 ahmed@iti.com
 omar@iti.com
(3 rows)

postgres=#
```

8- Write a query to calculate the difference in days between the current date, and each exam date.

```
postgres=# SELECT id, date, CURRENT_DATE - date AS days_difference
FROM exam;
 id |    date    | days_difference
----+------------+-----------------
  1 | 2025-04-29 |               1
  5 | 2025-04-30 |               0
  2 | 2025-04-25 |               5
(3 rows)

postgres=#
```

9- Write a query to check if each student's email address ends with '.com'.

```
postgres=# SELECT e_name, email,
CASE
WHEN RIGHT(email, 4) = '.com' THEN 'Yes'
ELSE 'No'
END
FROM student;
 e_name  |      email       | case
---------+------------------+------
 Mohamed | mohamed@iti.com  | Yes
 Ahmed   | ahmed@iti.com    | Yes
 Omar    | omar@iti.com     | Yes
(3 rows)
```

10- Display each exam date like ‘MM/DD/YYYY’.

```
postgres=# SELECT TO_CHAR(date, 'MM/DD/YYYY')
FROM exam;
  to_char
------------
 04/29/2025
 06/15/2025
 04/30/2025
(3 rows)

postgres=#
```